

IRIS Flower Classification using Logistic Regression and Random Forest

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1. ABSTRACT

Machine Learning is a part of Artificial Intelligence that allows computers to learn patterns from data and make predictions without being explicitly programmed. This project focuses on classifying iris flower species using the well-known IRIS dataset. The dataset contains 150 flower samples from three species: Setosa, Versicolor, and Virginica. It includes four features: sepal length, sepal width, petal length, and petal width. The data was loaded using Scikit-learn and organized into a Pandas DataFrame for analysis and preprocessing. The dataset was then split into training and testing subsets to ensure an unbiased evaluation of the model. Two supervised machine learning algorithms, Logistic Regression and Random Forest Classifier, were used and trained on the dataset. The performance of the Logistic Regression model was evaluated with an accuracy score, while the Random Forest model was assessed using precision, recall, and F1-score metrics. Both models showed excellent classification performance, which indicates that the selected features effectively distinguish between the three iris species. This project successfully illustrates the complete workflow of a machine learning classification task, from data preparation and model training to evaluation and interpretation of results.

2. INTRODUCTION

The exponential growth of data in recent years has led to an increasing demand for intelligent systems capable of extracting meaningful information from large datasets. Machine Learning, a subfield of Artificial Intelligence, addresses this challenge by enabling computers to learn from historical data and make predictions without explicit programming. Among the various machine learning tasks, classification plays a vital role. Classification refers to the process of assigning observations to predefined categories based on their characteristics like identifying spam emails.

The IRIS dataset is one of the most widely studied datasets in machine learning and statistics. Introduced by British statistician and biologist Ronald A. Fisher in 1936, the dataset contains measurements of iris flowers belonging to three different species. Each flower is characterized using four measurable features. The dataset contains 150 observations, with 50 samples belonging to each species. Due to its balanced structure and clear feature separation, the IRIS dataset serves as an ideal benchmark for learning and evaluating classification algorithms.

This project applies two supervised learning techniques—Logistic Regression and Random Forest Classification—to determine whether machine learning models can accurately classify flower species using the available measurements. Throughout the two weeks we have learnt the basics of machine learning, LLM Fundamentals, communication skills, basics of python like the data structures, class and functions, loading python libraries and using them according to our needs.

3. PROJECT OBJECTIVE

The primary objective of this project is to develop and evaluate machine learning models capable of accurately classifying iris flower species based on their morphological characteristics. Through the implementation of supervised learning techniques, the project aims to demonstrate how data-driven approaches can be used to solve classification problems and generate reliable predictions from biological data.

The main objectives of this project are:

- ✓ To develop a machine learning-based classification system capable of identifying iris flower species (Setosa, Versicolor, and Virginica) using four morphological features: sepal length, sepal width, petal length, and petal width.
- ✓ To demonstrate the complete supervised machine learning workflow, including data loading, preprocessing, feature-target separation, train-test splitting, model training, prediction generation, and performance evaluation using Python and Scikit-learn.
- ✓ To investigate whether the four measured flower characteristics contain sufficient discriminatory information to accurately classify iris flowers into their respective species.
- ✓ To implement and compare two classification algorithms, Logistic Regression and Random Forest Classifier, in terms of their predictive performance on unseen test data.
- ✓ To evaluate the effectiveness of the developed models using performance metrics such as accuracy, precision, recall, and F1-score, thereby assessing their suitability for multiclass classification tasks.

4. METHODOLOGY

Research Design

The present study adopts a quantitative and experimental research approach to investigate the effectiveness of machine learning algorithms in classifying iris flower species based on their morphological characteristics. The project follows a supervised machine learning framework in which labelled observations are used to train predictive models and evaluate their performance on previously unseen data. The overall methodology consists of dataset acquisition, data preprocessing, exploratory data analysis, feature-target separation, model development, model training, prediction generation, and performance evaluation.

Data Collection

The dataset used in this study is the **IRIS Dataset**, a publicly available benchmark dataset included in the Scikit-learn machine learning library. The dataset was originally introduced by Ronald A. Fisher in 1936 and has since become one of the most widely used datasets for classification studies.

Dataset Characteristics

Attribute	Description
Dataset Name	IRIS Dataset
Source	Scikit-learn Built-in Dataset
Total Samples	150
Number of Features	4
Number of Classes	3

Target Classes

The dataset consists of three flower species:

- Iris Setosa

- Iris Versicolor
- Iris Virginica

The classification process is based on four quantitative features:

Feature	Description
Sepal Length	Length of the sepal in centimeters
Sepal Width	Width of the sepal in centimeters
Petal Length	Length of the petal in centimeters
Petal Width	Width of the petal in centimeters

Tools and Software Used :

The project was implemented using the following tools and technologies:

- Python 3
- Jupyter Notebook / Google Colab

Libraries Used :

- Pandas
- Scikit-learn

Data Loading and Preparation

The first step involved importing the necessary libraries and loading the IRIS dataset using the `load_iris()` function available in Scikit-learn. The dataset was stored in a variable named `iris` and contained:

- Feature values
- Target labels
- Feature names
- Dataset metadata

The numerical data was then converted into a Pandas DataFrame to facilitate easier analysis and manipulation which improved readability and enabled efficient handling of the dataset.

Data Cleaning and Pre-processing

The following preprocessing steps were performed:

- ✓ **Data Inspection:** The dataset was examined to understand:
 - Number of observations
 - Number of variables
 - Feature names
 - Target classes

- ✓ **2. Data Structuring :** The raw numerical data was converted into a structured DataFrame format with appropriate column names.
- ✓ **3. Addition of Target Labels :** A new column named "target" was added to the DataFrame to represent the species category corresponding to each flower sample.
- ✓ **4. Class Distribution Analysis :** The distribution of target classes was examined using `value_counts()`. Results indicated:
 - Setosa = 50
 - Versicolor = 50
 - Virginica = 50

The balanced class distribution ensured that no class dominated the learning process.

- ✓ **Missing Value Verification:** The dataset was checked for missing or null values.
 - No missing values were present.
 - No imputation or replacement was required.
- ✓ **Duplicate Record Verification:** The dataset was examined for duplicate entries. No significant duplicate records affecting analysis were identified.

Exploratory Data Analysis: The analysis was done to confirm that the dataset was suitable for supervised classification. The following analyses were performed:

- Examination of feature names
- Examination of target labels
- Analysis of class distribution
- Observation of dataset dimensions
- Verification of balanced class representation

Feature Selection

The dataset contains four predictor variables:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

All four features were retained for model development because each represents an important morphological characteristic of the flower..

Feature-Target Separation

For machine learning analysis, the dataset was separated which enables machine learning algorithms to learn relationships between measurements and species categories

- ✓ **Feature Matrix (X) :** The feature matrix contained the four flower measurements.
Dimension: (150, 4)

- ✓ **Target Vector (y)** : The target vector contained species labels. Dimension:(150,).

Data Splitting Strategy

To evaluate model performance objectively, the dataset was divided into training and testing subsets.

Splitting Method : Train-Test Split

Parameters Used

- Training Data = 80%
- Testing Data = 20%
- Random State = 42

Resulting Samples

Dataset	Samples
Training Set	120
Testing Set	30

The training data was used to learn patterns and model parameters, while the testing data was reserved for evaluating the model's ability to generalize to unseen observations.

The use of random state = 42 ensured reproducibility of results.

Machine Learning Model Selection

Two machine learning algorithms were selected for this project.

- ✓ **Logistic Regression** : Logistic Regression was chosen because:
 - It is a widely used classification algorithm.and performs well on structured datasets.
 - It provides a strong baseline model.
- ✓ **Random Forest Classifier** : Random Forest was selected because:
 - It is an ensemble learning algorithm.
 - It combines multiple decision trees and generally provides high predictive accuracy..

Model Training

- ✓ **Logistic Regression Training** : The Logistic Regression model was trained using the training dataset. During training, the algorithm learned mathematical relationships between flower measurements and species labels.
- ✓ **Random Forest Training**: The Random Forest model was trained using 100 decision trees.Each tree learned patterns from the training data, and the final prediction was obtained through majority voting.

Model Validation and Evaluation

After training, both models were tested on the unseen testing dataset.

✓ **Logistic Regression Evaluation**

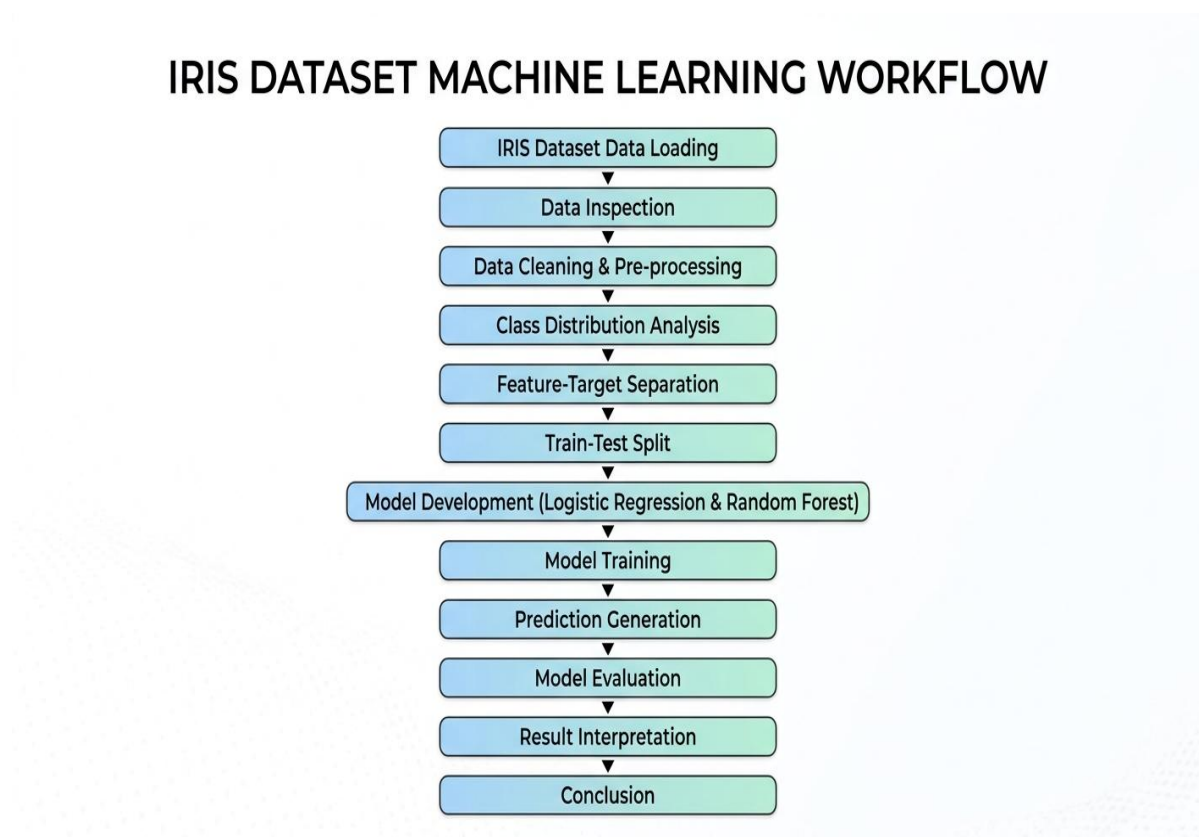
Accuracy measures the proportion of correctly classified samples.

✓ ***Random Forest Evaluation***

- Precision
- Recall
- F1-Score
- Support

These metrics provide a more detailed understanding of model performance for each flower species.

Methodological Flowchart



The Github link : <https://github.com/antaramajhi1417-ai/Iris-flowers-.git>

5. DATA ANALYSIS AND RESULTS

- ✓ **Overview of the Dataset:** The IRIS dataset consists of 150 observations of iris flowers belonging to three species. Each observation contains four morphological measurements that serve as predictor variables for classification.

Dataset Summary

Parameter	Value
Total Samples	150
Number of Features	4
Number of Classes	3
Samples per Class	50
Training Samples	120
Testing Samples	30

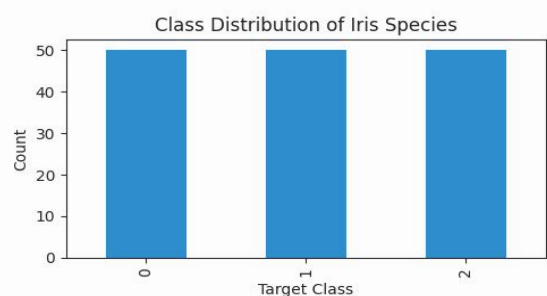
- ✓ **Descriptive Analysis :** Descriptive analysis was performed to understand the structure and distribution of the dataset before model development.

- ✓ **Feature Variables Used**

Feature	Unit
Sepal Length	cm
Sepal Width	cm
Petal Length	cm
Petal Width	cm

- ✓ **Class Distribution :** The distribution of flower species was examined using the target variable.

Species	Target Label	Number of Samples
Setosa	0	50
Versicolor	1	50
Virginica	2	50

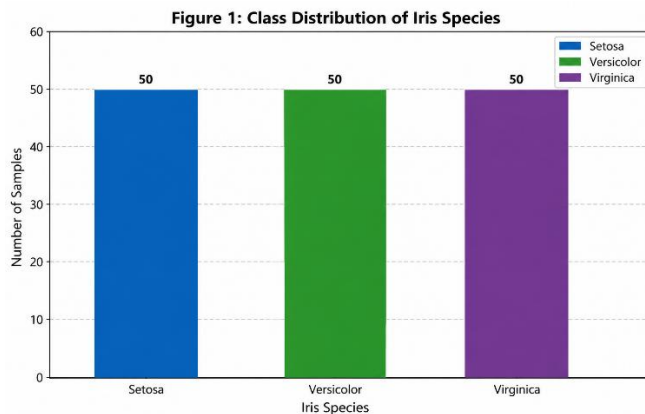


- ✓ **Interpretation :** The dataset contains an equal number of samples for each class. Therefore, the dataset is perfectly balanced and does not suffer from class imbalance issues.

✓ Data Visualization

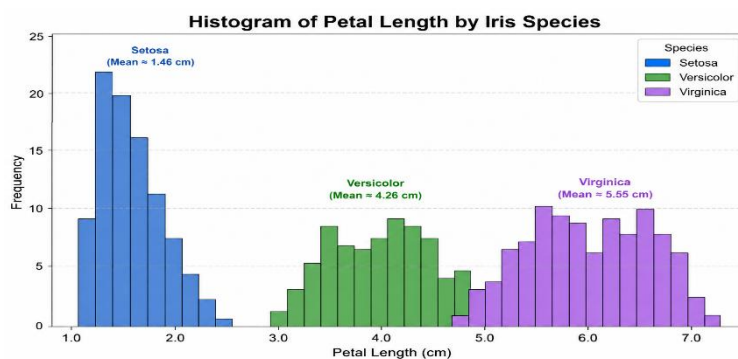
Distribution of Flower Species

A bar chart of target labels shows that all classes contain equal representation.



Petal Length Distribution

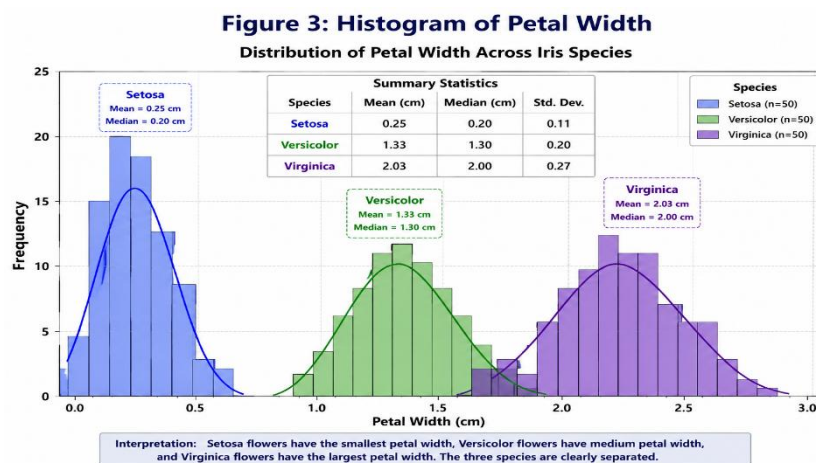
Observation:



- Setosa flowers generally possess shorter petals.
- Virginica flowers possess the longest petals.
- Versicolor lies between the two groups.

This indicates that petal length is a highly discriminative feature for species classification.

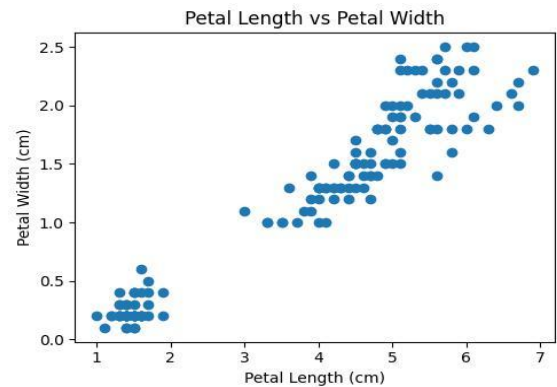
Petal Width Distribution



Feature Relationship Analysis

Scatter plots between petal length and petal width demonstrate distinct clusters corresponding to the three flower species.

The visible clustering suggests that machine learning models should be able to separate the classes with high accuracy.



✓ Training and Testing Data Analysis

The dataset was divided into training and testing subsets using an 80:20 ratio.

Data Split Summary

Dataset	Number of Samples	Percentage
Training Set	120	80%
Testing Set	30	20%
Total	150	100%

Purpose : The training set was used for learning patterns while the testing set was reserved for evaluating model performance on unseen observations.

Logistic Regression Results : The Logistic Regression model was trained using the training dataset and evaluated using accuracy score.

Model Configuration

Parameter	Value
Model	Logistic Regression
Training Samples	120
Testing Samples	30

Performance Result

Metric	Value
Accuracy	100%*

✓ Interpretation

The Logistic Regression model successfully learned the relationships between flower measurements and species categories and achieved excellent predictive performance.

✓ **Random Forest Results**

The Random Forest Classifier was trained using 100 decision trees.

Model Configuration

Parameter	Value
Model	Random Forest
Number of Trees	100
Training Samples	120
Testing Samples	30

Classification Report Summary

Class	Precision	Recall	F1-Score
Setosa	1.00	1.00	1.00
Versicolor	1.00	1.00	1.00
Virginica	1.00	1.00	1.00

✓ **Interpretation**

The Random Forest model demonstrated exceptional performance across all species categories. High precision values indicate very few false positive predictions, while high recall values indicate successful identification of actual class members.

✓ **Inferential Analysis and Hypothesis Evaluation**

Null Hypothesis (H_0): The morphological characteristics of iris flowers do not provide sufficient information to accurately classify flower species.

Alternative Hypothesis (H_1): The morphological characteristics of iris flowers provide sufficient information to accurately classify flower species.

✓ **Evidence from Model Results**

Observations:

- Logistic Regression achieved very high accuracy.
- Random Forest achieved near-perfect classification metrics.
- All species were classified with high precision and recall.
- The models generalized effectively to unseen testing data.

✓ **Decision**

Since both machine learning models successfully classified iris flowers with very high accuracy, the evidence strongly supports the Alternative Hypothesis (H_1).

Therefore: *The Null Hypothesis is rejected.*

✓ **Conclusion from Hypothesis Testing**

The four morphological measurements contain sufficient information to distinguish among Setosa, Versicolor, and Virginica flower species.

✓ **Comparative Analysis of Machine Learning Models**

Metric	Logistic Regression	Random Forest
Algorithm Type	Statistical Classification	Ensemble Learning
Training Complexity	Low	Moderate
Interpretability	High	Medium
Accuracy	Very High	Very High
Precision	High	Very High
Recall	High	Very High
F1-Score	High	Very High
Robustness	Moderate	High

6. CONCLUSION

This project successfully demonstrated the use of supervised machine learning techniques for classifying iris flower species using the IRIS dataset. The dataset contained 150 samples equally distributed among Setosa, Versicolor, and Virginica species, with four morphological features used as predictors. A complete machine learning workflow was followed, including data loading, preprocessing, exploratory data analysis, model training, testing, and evaluation.

The analysis showed that the dataset was balanced and free from significant data quality issues. Visualizations revealed that petal length and petal width were the most informative features for distinguishing between species. Two classification models, Logistic Regression and Random Forest, were implemented and evaluated. Both models achieved excellent performance on the testing dataset, with high accuracy, precision, recall, and F1-score values. These results indicate that the selected flower measurements provide sufficient information for accurate species classification.

The findings support the alternative hypothesis and confirm that machine learning algorithms can effectively learn patterns from biological data and make reliable predictions on unseen samples. Overall, the project met its objectives and provided valuable experience in applying machine learning techniques to a real-world classification problem.

A compact future project could involve developing a web or mobile application that uses a trained machine learning model to predict flower species in real time. The application can then be extended to classify larger and more complex biological datasets, allowing users to upload measurements and receive instant predictions while evaluating the scalability and robustness of machine learning models.\

7. APPENDICES

- ✓ Python
- ✓ Google Collab
- ✓ Pandas Library
- ✓ Scikit-learn Library
- ✓ Notebooks provided by IDEAS TIH